

A Novel Integrated Multi-Gb/s Optical Wireless Communication and Sub- μm Particle Sensing System for IoT Applications

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Abstract A proof-of-concept optical wireless integrated sensing and communication (OW-ISAC) system is proposed, demonstrating 0.5–10 μm particulate matter (PM) detection and 7.3 Gb/s optical wireless communication (OWC) at hard-decision forward error correction (HD-FEC) bit error ratio (BER), using a single laser diode (LD). ©2026 The Author(s)

Introduction

Optical wireless communication (OWC) based on intensity modulation and direct detection (IM/DD) offers a cost-effective and low-latency connectivity solution for Internet of things (IoT) and Industry 4.0 systems, such as smart buildings and industrial monitoring [1–4]. In addition, integrated sensing and communication (ISAC) enables every IoT link to act as a micro-sensing probe, creating a dense sensing fabric for seamless environmental monitoring including air quality monitoring [5–7]. In urban areas, pollution is a major public health concern. In fact, 94% of the urban population in European Union (EU) is exposed to particulate matter (PM) with aerodynamic diameters less than 2.5 μm (i.e., $\text{PM}_{2.5}$) at concentrations exceeding the guideline level recommended by the World Health Organization (WHO) [8]. In industrial environments, PM monitoring is also essential for process stability and quality control [9, 10].

To date, a few studies related to OWC systems have addressed particle monitoring. In [11], the authors demonstrated a fine particle monitoring system that transmits real-time $\text{PM}_{2.5}$ data from dust sensors reliably over indoor visible light communication (VLC) links based on direct current-biased optical orthogonal frequency division multiplexing (DCO-OFDM). In [12], the authors presented an indoor PM monitoring system for smart home applications, transmitting $\text{PM}_{2.5}$ and PM_{10} data over multi-hop VLC links using non-return-to-zero on-off keying (NRZ-OOK). This work puts forward and experimentally validates an optical wireless ISAC (OW-ISAC) system that enables joint sub- μm PM sensing and high-speed data transmission based on a single laser diode (LD). The proposed system is capable of detecting particles with diameters as small as 0.5 μm while achieving data rates of up to 7.3 Gb/s using DCO-OFDM with IM/DD. Since the same LD of the particle sensing platform is used for PM monitoring and data communication, it greatly reduces hardware complexity and facilitates seamless integration with IoT networks.

Theory and Experimental Setup

Fig. 1 shows the design of the proposed OW-ISAC system. Details of the system architecture and the experimental setup are discussed in the following. Fig. 1(a) depicts the architecture of the end-to-end OW-ISAC system. The particle sensing platform lying at the core of the system is composed of an infrared (IR) laser source with collimating optics and a sensing chamber connected to air pipes, allowing particles to flow through the chamber using an air pump. A filter is used to remove excess particles from the air before it is pumped out of the chamber. When particles enter the sensing chamber, they are illuminated by the laser beam, and the resulting scattered light is reflected toward the sensing photodiode (PD) by a concave mirror for detection. The optical power of the scattered light can be expressed as $P_s = IQ_s\pi r_p^2$, where I is the incident optical intensity, Q_s is the Mie scattering efficiency, and r_p is the particle radius [13]. The sensing PD generates an electrical photocurrent signal whose amplitude is proportional to the received optical power of the scattered light and hence to the particle size. The detected photocurrent is amplified and converted into a voltage signal by a transimpedance amplifier (TIA), which is subsequently digitized by an analog-to-digital converter (ADC). The resulting digital pulses are processed on a laptop computer to detect peak values, count and classify particles into different PM size groups based on predefined amplitude thresholds. The measured PM counts are used to estimate the PM mass concentration for each size group assuming that the particles are spherical. The PM mass concentration in the i th size group before calibration is given by $m_i = \frac{\pi}{6} N_i d_i^3 \rho_{\text{eff}}$, where N_i is the number of detected particles and d_i is the representative particle diameter for the i th size group, and ρ_{eff} is the effective particle density [14]. A calibration factor of k_{cal} , obtained from the reference measurements, is then applied to compensate for the discrepancies between the estimated and reference PM values. The calibrated PM concentration

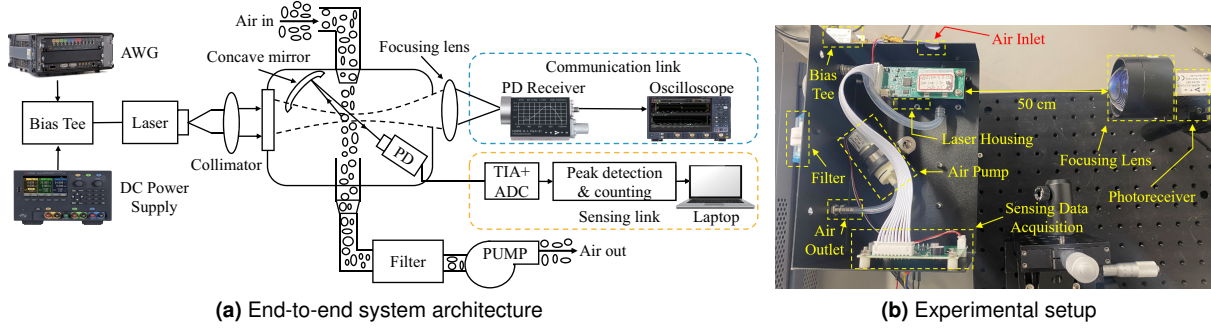


Fig. 1: Laser-based OW-ISAC system configuration for joint sub- μm particle sensing and multi-Gb/s data transmission.

is calculated as $\text{PM}_\ell = k_{\text{cal}} \sum_{d_i \leq \ell} m_i$, where ℓ denotes the upper particle diameter limit for PM_1 , $\text{PM}_{2.5}$, or PM_{10} . Also, the calibrated total suspended particles (TSP) concentration is estimated as $\text{TSP} = k_{\text{cal}} \sum_i m_i$.

Fig. 1(b) shows the experimental setup of the OW-ISAC system. The same 785 nm index-guided aluminium gallium arsenide (AlGaAs) LD (SONY SLD238VL) of the sensing platform is used for the communication link. The measured characteristics of this LD are presented in Fig. 2. The light-current-voltage (LIV) curves were measured at room temperature with the DC bias current, I_{DC} , ranging between 30 mA and 140 mA, as shown in Fig. 2(a). The manufacturer has recommended the upper limit of I_{DC} to avoid damaging the LD. It provides a maximum output power of about 100 mW and a linear dynamic range between forward currents of 50 mA and 140 mA for DCO-OFDM modulation. The far field (FF) beam profile was measured at 5 cm from the laser output at a forward current of 90 mA, exhibiting an elliptical profile, as shown in Fig. 2(b). In order to establish an IM/DD OWC link with high spectral efficiency, the LD is directly modulated using DCO-OFDM with adaptive M -ary quadrature amplitude modulation (QAM). The modulation and demodulation of the OFDM signal are performed using N_{FFT} -point inverse fast Fourier transform (IFFT) and FFT operations, based on offline digital signal processing (DSP). At the transmitter, after root raised cosine pulse shaping, the arbitrary waveform generator (AWG) (Keysight M8195A) converts the digital waveform into a wideband analog signal with a bandwidth of 4 GHz. Upon adding a DC bias current with the bias tee (Mini-Circuits ZFBT-4R2G), the output signal drives the forward current of the LD. The modulated laser beam is propagated through free space toward the receiver (FEMTO HSPR-X-I-1G4-SI) over a link distance of 50 cm. This photoreceiver is made of a silicon positive-intrinsic-negative (PIN) PD with an active area diameter of 400 μm , a bandwidth of 1.4 GHz, and a responsivity of 0.5 A/W at 785 nm. The output signal of the photoreceiver is digitized using the sampling oscilloscope (Keysight UXR0104B). The rest of DSP operations including FFT, single-tap equalization and QAM demapping

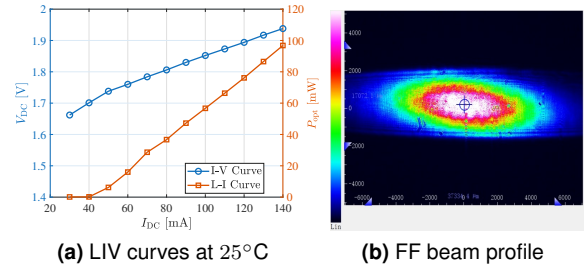


Fig. 2: Measured characteristics of the 785 nm AlGaAs LD.

are executed in MATLAB. Each OWC experiment involves a total of 450 OFDM data frames, of which 150 pilot frames are used for channel estimation to perform adaptive bit and power loading. The data rate of the DCO-OFDM-based OWC link is [15]:

$$R = \frac{2B}{N_{\text{FFT}} + N_{\text{CP}}} \sum_{k=1}^{N_{\text{SC}}} b_k, \quad (1)$$

where $b_k = \log_2 M_k$ is the number of bits allocated to the k th subcarrier and M_k is the corresponding QAM constellation size, B is the signal bandwidth, N_{SC} is the number of data-carrying subcarriers, and N_{CP} is the cyclic prefix (CP) length. In this work, $B = 4$ GHz, $N_{\text{FFT}} = 1024$, and $N_{\text{CP}} = 20$.

Performance Measurement Results

Fig. 3 shows complete measurement results to evaluate the performance of the OW-ISAC system. Fig. 3(a) demonstrates how the sensing platform detects particles by generating a voltage pulse with each particle passing through the chamber. It uses voltage thresholds at 0.2 V and 1.0 V to determine whether the PM size is in the size range 1–2.5 μm or 2.5–10 μm , respectively. During the measurement period, pulses with low peak amplitudes below 1.0 V appear more frequently, while fewer pulses with larger amplitudes are observed, which indicates that smaller particles under 2.5 μm dominate the measurement. Following the same approach, the sensing platform computes the number of particles (i.e., N_i) counted in five different PM size groups from 0.5 μm to 10 μm in a given period of time, as shown in Table 1 for a 5 s period. These values are converted into the corresponding PM and TSP concentration values based on the discussion in the previous section. Fig. 3(b) shows

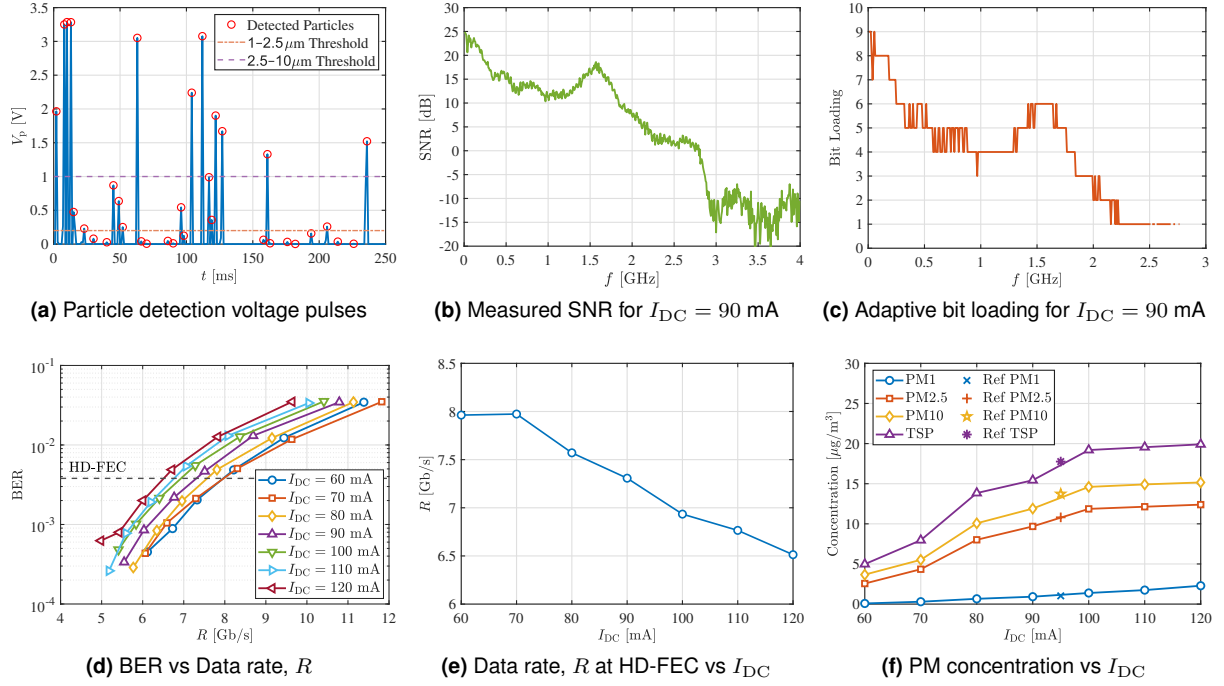


Fig. 3: Measured performance of the OW-ISAC system.

Tab. 1: Particle counts in different size groups over 5 seconds.

t [s]	PM diameter [μm]				
	0.5	1.0	2.5	5.0	10.0
0	240	165	78	24	12
1	228	186	56	18	5
2	205	143	55	15	8
3	254	182	47	19	5
4	188	136	42	11	2

the measured signal-to-noise ratio (SNR) of the OWC link versus frequency for $I_{DC} = 90$ mA. The peak-to-peak amplitude of the transmitted OFDM signal is set to 0.4 V. At low frequencies, the SNR is in the 20–25 dB range, and it reaches a local peak value of about 18 dB at 1.6 GHz, then exhibits a decreasing trend until it crosses 0 dB at 2.8 GHz, which specifies the effective bandwidth. Fig. 3(c) shows the corresponding adaptive bit loading results, where up to 9 bits are allocated to low-frequency subcarriers, dropping down to 1 bit in the high frequency range of 2.2 GHz to 2.8 GHz.

Fig. 3(d) shows the bit error ratio (BER) versus data rate performance for different values of I_{DC} . Based on Fig. 3(d), Fig. 3(e) shows the achievable data rate, R , at the hard-decision forward error correction (HD-FEC) BER of 3.8×10^{-3} as a function of I_{DC} in the range of 60–120 mA. As I_{DC} increases, R decreases from about 8 Gb/s to 6.5 Gb/s. This is because by increasing I_{DC} while keeping the peak-to-peak amplitude of the signal the same, the signal variations approach the upper end of the LIV curves of the LD, as shown in Fig 2(a), reducing the SNR and thus the achievable data rate due to nonlinearity. On the other hand, Fig. 3(f) shows the measured PM mass concentration in terms of PM₁, PM_{2.5}, PM₁₀,

and TSP with respect to I_{DC} , including their calibrated values at $I_{DC} = 95$ mA for optimal performance, which are used as a reference for comparison. It can be seen that all four parameters follow an increasing trend with I_{DC} , due to the fact that when I_{DC} and hence the output power of the LD increases, the particle detection sensitivity goes up. Comparing Fig. 3(e) with Fig. 3(f), there is evidently a tradeoff between the achievable rate performance and the particle sensing performance of the OW-ISAC system. For $I_{DC} = 90$ mA, the measured values of PM₁, PM_{2.5}, PM₁₀, and TSP are approximately 0.92, 9.68, 11.90, and 15.43 $\mu\text{g}/\text{m}^3$, respectively, with relative errors of 10.2%, 10.4%, 13.3%, and 13.1% with respect to the reference values. These error values are below 15%, fulfilling the size resolution requirement of the ISO 21501-4 standard [16]. As a result, $I_{DC} = 90$ mA strikes a balance between the achievable data rate and the particle detection sensitivity of the OW-ISAC system. In this case, the system achieves a data rate of 7.3 Gb/s at the target HD-FEC BER.

Conclusions

A laser-based OW-ISAC system for joint PM monitoring and OWC using a single LD was experimentally demonstrated, confirming that there is a tradeoff between the achievable data rate and the particle detection performance with respect to the DC bias current of the LD. A balanced performance is achieved for $I_{DC} = 90$ mA. In this case, errors of the measured values of PM₁, PM_{2.5}, PM₁₀, and TSP relative to their calibrated reference values are within the acceptable limit of 15%, while the OWC link reaches 7.3 Gb/s at HD-FEC BER.

Acknowledgement

This work was supported by the Engineering and Physical Sciences Research Council (EPSRC) Centre for Doctoral Training (CDT) in Photonic and Electronic Systems (PES) and the EPSRC project ‘Future Communications Hub in All-Spectrum Connectivity (HASC)’ under grant EP/Y037197/1.

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